Package ‘camRa’

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Type Package

Title Provides Utilities for Managing Camera Trap Data

Version 0.0.1.9000

Description Provides utility functions aimed at making it easier to pull in and edit data related to camera traps. This includes manipulating detection files from MegaDetector and SpeciesNet, extracting data from/editing imagesas, and downloading LILA datasets without needing to learn how to use AWS among other things.

License GPL (>= 3)

URL https://oxyppgyn.github.io/camRa, https://https://github.com/oxyppgyn/camRa

BugReports https://https://github.com/oxyppgyn/camRa/issues

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 8.0.0

Imports jsonlite,aws.s3, DBI, RSQLite, magick, tesseract

Depends R (>= 3.5)

LazyData true

Suggests knitr, rmarkdown, dplyr, stringr, ggplot2

VignetteBuilder knitr

Collate 'common.R'
'Image.R'
'LILA.R'
'MegaDetector.R'
'Timelapse.R'
'data-documentation.R'

Contents

`convert_bbox` *Convert between BBox Formats*

Description

Converts bounding box data between three common formats: xyxy, xywh, and centerwh. WARNING: This function is experimental, math behind some conversions may be wrong and has not yet been validated.

Usage

```
convert_bbox(
  bbox,
  from,
  to,
  image = NULL,
  img_width = NULL,
  img_height = NULL,
  value_type = "absolute"
)
```

Arguments

`bbox` character vector. Bounding box values.

`from, to:`
- `from` character vector. Bounding box format to convert from.
- `to` character vector. Bounding box format to convert to.
- Options: "xyxy", "xywh", and "centerwh"

`image` character or magick-image. The image to extract width and height from. Only used if converting between value types (`value_type` is opposite of the provided bounding box). Value type of the input bounding box is inferred. Mutually exclusive with `img_width` and `img_height`.

`img_width, img_height`
numeric. The width and height of the image this bounding box belongs to. Only used if converting between value types (`value_type` is opposite of the provided bounding box). Value type of the input bounding box is inferred. Mutually exclusive mutually exclusive with `image`.

`value_type` character. Type of values to use for the converted bounding box measures. Either "absolute" for pixel values or "relative" for scaled values.

Value

a numberic vector.

`ena24detection_subset` *Random Subset of ena24detection files*

Description

A random subset of 500 image files from the ena24detection dataset to be used for testing.

Usage

```
ena24detection_subset
```

Format

An object of class `data.frame` with 500 rows and 4 columns.

`img_apply` *Apply a Summary function to an Image*

Description

Applies a function to the extracted numeric matrix of values behind an image and returns the result. Functions are applied using `apply()` across all channels.

Usage

```
img_apply(image, fun, file = NULL, overwrite = FALSE, bbox = NULL, ...)
```

Arguments

`image`	character or magick-image. Image to extract information from.
`fun`	function reference. The function to be applied.
`bbox`	numeric vector. Bounding box values. Formatted as xmin, ymin, xmax, ymax. Use `convert_bbox()` to convert from other formats.
`...`	Additional arguments passed to `fun`.

Value

varies based on function applied.

img_difference — *Calculate Differences Between Two Images*

Description

Calculates the absolute pixel difference between two images.

Usage

```
img_difference(
  image1,
  image2,
  threshold = NULL,
  file = NULL,
  overwrite = FALSE,
  bbox = NULL
)
```

Arguments

`image1, image2`	character or magick-image. Images to be compared.
`threshold`	numeric. The image threshold that should be applied between 0 and 1. All values below the threshold will be converted to 0.
`file`	character. File path to the output image.
`bbox`	numeric vector. Bounding box values. Formatted as xmin, ymin, xmax, ymax. Use `convert_bbox()` to convert from other formats.

Value

a magick image.

img_extract_text — *Extract Text from Image Using OCR*

Description

Extracts text from an image using optical character recognition (OCR). OCR is notably unreliable for some camera types, text extraction should be used as a last resort and may not be accurate. Check the results of test images before trusting OCR text outputs.

Usage

```
img_extract_text(image, bbox = NULL, file = NULL, overwrite = FALSE, ...)
```

Arguments

`image`	character or magick-image. The image to extract text from.
`bbox`	numeric vector. Bounding box values. Formatted as xmin, ymin, xmax, ymax. Use `convert_bbox()` to convert from other formats. using absolute pixel values (Top-Left Bottom-Right format). Positions are measured from the top left corder of the image. Use `NULL` to get text from the entire image.
`...`	additional arguments passed to OCR engine as `tesseract::tesseract()`.

Value

a character.

`img_luminosity` *Calculate Luminosity/Brightness of an Image*

Description

Extracts luminance information from a specified colorspace and returns the appropriate channel.

Usage

```
img_luminosity(
  image,
  method = "grayscale",
  file = NULL,
  overwrite = FALSE,
  bbox = NULL
)
```

Arguments

`image`	character or magick-image. Image to extract information from.
`method`	character. The method used to calculate luminosity/brightness. Options are "greyscale" or "grayscale" to get a greyscale version of an RGB image, "LAB" to extract the lightness (L) channel from a LAB colorspace, "HSV" to extract the value (V) channel from a HVS colorspace, "YCbCr" for the luminance channel (Y) from a YCbCr color space, or "YIQ" for the luma (Y) channel in YIQ colorspace.
`file`	character. File path to the output image.
`bbox`	numeric vector. Bounding box values. Formatted as xmin, ymin, xmax, ymax. Use `convert_bbox()` to convert from other formats.

Value

a magick image.

`LILA_download` *Download LILA Image Files*

Description

Downloads subsets of image files from AWS S3 buckets associated with LILA datasets.

Usage

```
LILA_download(dataset, files, dir, quiet = FALSE)
```

Arguments

`dataset`	character. Name of the dataset to download images from. Can be either the short name or standard name. See `LILA_list_datasets()`.
`files`	character or character vector. Partial path(s) or file name(s) of all images to be downloaded. Files may have partial paths included here (such as `folder1/imgname.JPG`) depending on how buckets are organized. If all files are stored in the same "folder" through S3, you only need the file name with its extension. File names are typically available from JSON metadata files hosted on the LILA website or via `LILA_list_files()`.
`dir`	character. Directory where images will be saved.
`quiet`	logical. If information on file downloads should be printed to the console.

Value

`NULL`

Examples

```
#Download example subset files
LILA_download_files(
  dataset = "ena24detection",
  files = ena24detection_subset$file,
  dir = getwd()
)
```

`LILA_list_datasets` *Get a List of Datasets from LILA BC*

Description

Provides a dataframe with names (long and short) used for LILA datasets. Can be used to find valid values for other LILA functions.

Usage

```
LILA_list_datasets(quiet = FALSE, only_nonzip = TRUE)
```

Arguments

`quiet`	logical. If warnings should be shown.
`only_nonzip`	logical. If the returned datasets should be only ones available file-by-file (not exclusively as compressed files).

Value

A data frame with two columns: `DatasetName` and `DatasetShortName`.

`LILA_list_files` *List Files in LILA Datasets*

Description

Lists all files present in LILA datasets using AWS.

Usage

```
LILA_list_files(dataset)
```

Arguments

`dataset`	character. Name of the dataset to download images from. Can be either the short name or standard name. See `LILA_list_datasets()` for options.

Value

A character vector of file names with partial paths (if applicable) of files present in LILA datasets.

`megadet_filter` *Filter MegaDetector/SpeciesNet JSON Files by Image*

Description

Filters MegaDetector/SpeciesNet JSON files to only data for images in the directory specified.

Usage

```
megadet_filter(
  dir,
  json,
  file = NULL,
  overwrite = FALSE,
  validate_json = getOption("camRa.validate_json", default = TRUE)
)
```

Arguments

`dir`	character. Directory to be used to filter the JSON data. This folder should not be within the relative path of the `file` JSON tag, but should directly contain any folders that appear there.
`json`	character vector or nested list object from `jsonlite::read_json()`. The JSON file or loaded JSON data to filter on.
`file`	character. File to write JSON data to. Use `NA` to skip writing to a file.
`validate_json`	boolean. If JSON data formatted as nested lists should be validated. This can prevent unexpected errors if the parameter is a list, but not JSON but may increase runtime.
`overwrite.`	logical. If overwriting files is allowed.

Value

a nested list object with filtered JSON data.

`megadet_flatten` *Flatten MegaDetector/SpeciesNet JSON Detection Data*

Description

Flattens MegaDetector/SpeciesNet JSON files to a table format. Data is split by detection, with images potentially having more than one row of data if more than one detection was found. Images with no detections will have only one record here, with no detection (bbox) information.

Usage

```
megadet_flatten(
  json,
  map_names = TRUE,
  validate_json = getOption("camRa.validate_json", default = TRUE)
)
```

Arguments

`json`	character vector or nested list object from `jsonlite::read_json()`. The JSON file or loaded JSON data to filter on.
`map_names`	boolean. If MegaDetector and SpeciesNet names should be mapped from numeric values to the values provided in the `detection_categories` and `classification_categorie` tags. These are typically gross (human, animal, vehicle) and taxonomic classifications respectively.
`validate_json`	boolean. If JSON data formatted as nested lists should be validated. This can prevent unexpected errors if the parameter is a list, but not JSON but may increase runtime.

Value

a dataframe with all data from the `images` tag of the JSON file.

megadet_get_counts	*Count Number of Instances of MegaDetector/SpeciesNet Categories in Images*

Description

Counts instances of each class for MegaDetector/SpeciesNet across images in JSON data.

Usage

```
megadet_get_counts(
  json,
  type = "detection",
  map_names = TRUE,
  validate_json = getOption("camRa.validate_json", default = TRUE)
)
```

Arguments

`json`	character vector or nested list object from `jsonlite::read_json()`. The JSON file or loaded JSON data to filter on.
`type`	character. "detection" if counts should be done from MegaDetector's gross categories or "classification" for counts from SpeciesNet's species-level classifications.
`map_names`	boolean. If MegaDetector and SpeciesNet names should be mapped from numeric values to the values provided in the `detection_categories` and `classification_categorie` tags. These are typically gross (human, animal, vehicle) and taxonomic classifications respectively.
`validate_json`	boolean. If JSON data formatted as nested lists should be validated. This can prevent unexpected errors if the parameter is a list, but not JSON but may increase runtime.

Value

a dataframe with a category and count column.

megadet_get_info	*Get Values from MegaDetector/SpeciesNet JSON Files*

Description

Grabs values from MegaDetector/SpeciesNet JSON files based on provided keys.

Usage

```
megadet_get_info(
  json,
  key = "info",
  validate_json = getOption("camRa.validate_json", default = TRUE)
)
```

Arguments

`json`	character vector or nested list object from `jsonlite::read_json()`. The JSON file or loaded JSON data to filter on.
`key`	character, character vector, list, or `NULL`. Key(s) and/or indices to filter on in the JSON data. For root keys, provide a character of the key name. For higher levels, provide a vector or list of keys in the order they are nested. If `NULL` is passed as a key, the entire JSON will be returned.
`validate_json`	boolean. If JSON data formatted as nested lists should be validated. This can prevent unexpected errors if the parameter is a list, but not JSON but may increase runtime.

Value

dependent on keys used.

Examples

```
## Not run:
#Get data for first images in file
megadet_get_info(
  json = "classifications.json",
  key = list("images", 1)
)

#Get name of detector used
megadet_get_info(
  json = "classifications.json",
  key = c("info", "detector")
)

#Load entire JSON from file
megadet_get_info(
  json = "classifications.json",
  key = NULL
)

## End(Not run)
```

`specnet_reclassify` *Update Classification Information in SpeciesNet JSON Data*

Description

Reclassifies classification categories in SpeciesNet JSON data with new species values, including merging categories into one. This function is best used with a data frame mapping old values to new ones. A list of current classification names can be pulled using `megadet_get_info()` with key = `"classification_categories"` to build this table.

Usage

```
specnet_reclassify(
  json,
  values_from,
  values_to,
  values_description = NULL,
  file = NULL,
  overwrite = FALSE,
  validate_json = getOption("camRa.validate_json", default = TRUE)
)
```

Arguments

`json`	character vector or nested list object from `jsonlite::read_json()`. The JSON file or loaded JSON data to filter on.
`values_from`	character vector. A vector of classification names currently in the JSON to be updated. This must be unique values and include all names present in the file.
`values_to`	character vector. A vector of classification names to reassign current classification names as. Multiple copies of the same name can be present here to map values as many-to-one.
`values_description`	character vector. A vector of classification descriptions to reassign current classification descriptions as. This must have unique values to map onto `values_from`. Use `NA` to skip updating descriptions and instead remove them from the file.
`file`	character. File to write JSON data to. Use `NA` to skip writing to a file.
`overwrite`	logical. If overwriting files is allowed.
`validate_json`	boolean. If JSON data formatted as nested lists should be validated. This can prevent unexpected errors if the parameter is a list, but not JSON but may increase runtime.

Value

a nested list object representing JSON data.

Examples

```
## Not run:
#CSV with columns for old values, new values, and new descriptions columns
value_map <- read.csv("mapped_values.csv")

specnet_reclassify(
  json = "classifications.json",
  values_from = value_map$old_value,
  values_to = value_map$new_value,
  values_description = value_map$new_description
)

## End(Not run)
```

tl_ddb_extract *Get Data Table from Timelapse Database*

Description

Pulls the data table from a .ddb file used by the Timelapse program. Similar to the export to CSV option in the GUI.

Usage

```
tl_ddb_extract(file)
```

Arguments

file	character. File path to a timelapse database file (.ddb).

Value

a data frame.

tl_ddb_insert *Insert New Values for a Timelapse Database Column*

Description

Inserts a vector of values into a column in the data table of a Timelapse DDB file, overwriting existing data. Best used to insert data that cannot be obtained within Timelapse form metadata, such as OCR extracted temperature values.

Usage

```
tl_ddb_insert(file, col, values, ids)
```

Arguments

file	character. File path to a timelapse database file (.ddb).
col	character. Name of the column you want to update in the database file.
values	vector. A vector of values to insert into the database table. The class of this vector must align with the class of the database column.
ids	numeric vector. A vector of values with the IDs used in the table. `tl_ddb_extract()` can be used to extract this information and join with your datasets.

Value

NULL

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